

AYMANE AIT DADS

Machine Learning Computer Vision Intern | Object Detection · Image Classification · Deep Learning
Aymane.Ait-dads@eurecom.fr | +33 7 60 92 50 93 | linkedin.com/in/aymane-ait-dads | Sophia Antipolis, France

PROFESSIONAL SUMMARY

Ranked 1st in the EURECOM class leaderboard on an international AI-Generated Image Detection benchmark (0.791 AUC) by fine-tuning a CLIP ViT-L/14 backbone (428M parameters) across 250K images and 25+ generator types. Concurrently placing 17th / 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by designing an end-to-end LoRA SFT pipeline on a 30B-parameter model - demonstrating production-grade ML engineering across both computer vision and large-scale model training. Brings hands-on experience with PyTorch, convolutional and transformer-based architectures (CNN, DenseNet, ViT), ensemble methods, and ablation-driven experimentation directly applicable to Syntiant's detection pipeline research. Aims to expand computer vision detection capabilities through algorithm exploration and framework diversification within Syntiant's high-performance embedded AI systems.

CORE COMPETENCIES

Computer Vision Detection | Image Classification | Deep Learning Frameworks | Algorithm Exploration | Convolutional Neural Networks | Model Fine-Tuning | Ablation Studies | Production Pipeline Development

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Engineered a **Random Forest + K-Means classification pipeline** on 100K+ fiber-optic network records to detect quality degradation across thousands of network nodes.
- Developed an automated clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team.
- Reduced manual network analysis time by **~60%** through automated feature engineering, model evaluation, and reproducible Python-based reporting workflows.
- Delivered stakeholder presentations translating **model outputs** into actionable maintenance recommendations using Power BI dashboards for non-technical audiences.

PROJECTS

AI-Generated Image Detection - CLIP ViT Fine-Tuning

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in EURECOM class** (private Kaggle leaderboard, 0.791 AUC) by fine-tuning a CLIP ViT-L/14 backbone (428M params) with a custom MLP head trained on 250K images spanning 25+ generator types; also submitted to the **NTIRE 2026 @ CVPR CodaBench** international benchmark.
- Designed dual learning-rate scheduling (2e-6 CLIP backbone, 5e-4 MLP head), 10-view TTA (flips, crops, blur, resizes), and a **3-model ensemble** weighted by validation AUC, discovering that 1-epoch training (0.793 AUC) outperforms 4-epoch (0.767 AUC) for out-of-distribution generalization.

PyTorch · OpenCLIP · ViT · FP16 · Kaggle GPU

Aerial Cactus Detection - CNN / DenseNet / Transformer Benchmark

EURECOM Computer Vision · 2025 · 99.8% Accuracy

- Benchmarked CNN, **DenseNet121**, and hybrid CNN-Transformer architectures on 17,500+ drone images for endangered-species classification, achieving **99.8% accuracy, F1 = 0.999, and ROC AUC = 1.0**.
- Conducted systematic **algorithm exploration** across three distinct convolutional and transformer-based detection strategies to identify the optimal architecture for high-precision aerial image classification.

PyTorch · DenseNet121 · CNN · Scikit-learn · Matplotlib

NVIDIA Nemotron Model Reasoning Challenge - Kaggle (In Progress)

Kaggle · June 2026 · \$106K+ Prize Pool

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context).
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers for verified chain-of-thought data.

PyTorch · Unsloth · PEFT · vLLM · Triton · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM
Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Image Security, NLP, Cloud Computing, Statistics

Sep 2023 - Present

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

Computer Vision & Deep Learning: PyTorch | CLIP ViT | CNN | DenseNet | Image Classification | Object Detection | FP16/BF16 Mixed Precision

ML Frameworks & Engineering: Scikit-learn | Hugging Face Transformers | LoRA / PEFT | TRL / SFTTrainer | Unsloth | vLLM | Ablation Studies

Programming & Data Tools: Python | SQL | Pandas | NumPy | Matplotlib | Power BI | Git | Linux